INTRODUCTION TO INFORMATION SECURITY

WEEK 4

PLAYFAIR CIPHERS

An Introduction to Cryptography

Playfair ciphers

- Used in Dorothy L. Sayers' 1932 mystery novel *Have His Carcase*
- Marketing beats technology?
  - Invented by Charles Wheatstone
  - Lyon Playfair, a Scottish Baron, promoted it
  - Who got the glory?

Playfair Cipher

★ Multiple letter encryption cipher.

★ Digrams.

★ 5 x 5 matrix constructed using a keyword (Ex: Monarchy)

Playfair Cipher

- ★ Multiple letter encryption cipher.
- ★ Digrams.
- ★ 5 x 5 matrix constructed using a keyword (Ex: Monarchy)

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

PLAYFAIR CIPHERS

The Algorithm consists of 2 steps:

1. Generate the key Square(5×5):

1. The key square is a 5×5 grid of alphabets that acts as the key for encrypting the plaintext. Each of the 25 alphabets must be unique and one letter of the alphabet (usually J) is omitted from the table (as the table can hold only 25 alphabets). If the plaintext contains J, then it is replaced by I.

2. The initial alphabets in the key square are the unique alphabets of the key in the order in which they appear followed by the remaining letters of the alphabet in order.

Playfair ciphers

$$\begin{bmatrix} p & l & a & y & f \\ i & r & b & c & d \\ e & g & h & k & m \\ n & o & q & s & t \\ u & v & w & x & z \end{bmatrix}$$

- Matrix-based block cipher used in WWI

In a 5x5 matrix, write the letters of the word “playfair” (for example) without dups, and fill in with other letters of the alphabet, except I,J

USED INTERCHANGEABLY.

PLAYFAIR CIPHERS

2. Algorithm to encrypt the plain text: The plaintext is split into pairs of two letters (digraphs). If there is an odd number of letters, a Z is added to the last letter.

For example:

PlainText: "instruments"
After Split: 'in' 'st' 'ru' 'me' 'nt' 'sz'

PLAYFAIR CIPHERS

1. Pair cannot be made with same letter. Break the letter in single and add a bogus letter to the previous letter.

   Plain Text: “hello”

   After Split: ‘he’ ‘lx’ ‘lo’

   Here **‘x’** is the bogus letter.

2. If the letter is standing alone in the process of pairing, then add an extra bogus letter with the alone letter

   Plain Text: “helloe”

   AfterSplit: ‘he’ ‘lx’ ‘lo’ ‘ez’

   - Here **‘z’** is the bogus letter.

PRACTICE

- Make diagrams for
  - Attack
  - Neso academy
  - balloon

RULES FOR ENCRYPTION

if both the letters are in the same column: Take the letter below each one (going back to the top if at the bottom).

- For example:
- Diagraph: "me"
- Encrypted Text: cl
- Encryption:
- m -> c
- e -> l

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

RULES FOR ENCRYPTION

If both the letters are in the same row: Take the letter to the right of each one (going back to the leftmost if at the rightmost position).

- For example:
- Diagraph: "st"
- Encrypted Text: tl
- Encryption:
- s -> t
- t -> l

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

RULES FOR ENCRYPTION

If neither of the above rules is true: Form a rectangle with the two letters and take the letters on the horizontal opposite corner of the rectangle.

- For example:

- Diagraph: "nt"
- Encrypted Text: rq
- Encryption:
- n -> r
- t -> q

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Rules for encryption using Playfair Cipher

1. Digrams.
2. Repeating Letters - Filler letter.
3. Same Column | ↓ | Wrap around.
4. Same row | → | Wrap around.
5. Rectangle | ⇆ | Swap

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

PLAIN TEXT: "INSTRUMENTSZ"
ENCRYPTED TEXT: GATLMZCLRQTX

- Encryption:
- i -> g
- n -> a
- s -> t
- t -> l
- r -> m

u -> z

m -> c

e -> l

n -> r

t -> q

s -> t

z -> x

PLAIN TEXT: "INSTRUMENTSZ"
ENCRYPTED TEXT: GATLMZCLRQTX

in:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

st:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

ru:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

me:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

nt:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

sz:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z